

HR Attrition Analysis

Key Insights — Bangalore Office Attrition Investigation

44.8%

Bangalore's attrition rate vs. 27–31% across other offices

The Main Issue

Employees in the Bangalore office were leaving the company at a significantly higher rate than every other location — nearly 1 in 2 employees, compared to roughly 1 in 3-4 elsewhere. Leadership needed to know what was driving this and whether at-risk employees could be identified before they resigned, using 201,000 cleaned employee records spanning demographics, compensation, performance, and tenure.

Key Insights

1. The problem was confirmed genuine, not a data artifact

Before drawing any conclusions, location data (originally fragmented across 5+ inconsistent spellings) was standardized and re-verified. Bangalore's elevated attrition rate held after cleaning, confirming this was a real business problem rather than a reporting error.

2. Five confirmed drivers, ranked by strength

Driver	Effect Size	Comparison
Job satisfaction	~13 pp	Low satisfaction: ~52% vs High: ~39–40%
Overtime	~9 pp	Overtime: 50.9% vs No overtime: 41.9%
Salary hike %	~8 pp	Low hike (<6%): 51.4% vs Normal/High: 43.4%
Commute distance	~7.4 pp	Long (>25km): 51.3% vs Short: 43.9%
Tenure	~5.3 pp	Under 1 year: 49.3% vs 1+ years: 44.0%

3. Job satisfaction is the single highest-leverage factor

This conclusion was reached two independent ways: through direct rate comparison (the largest gap of any factor tested), and independently confirmed by a machine learning model, where satisfaction score carried more than double the predictive weight of any other individual factor. Agreement across two different methods makes this a high-confidence finding.

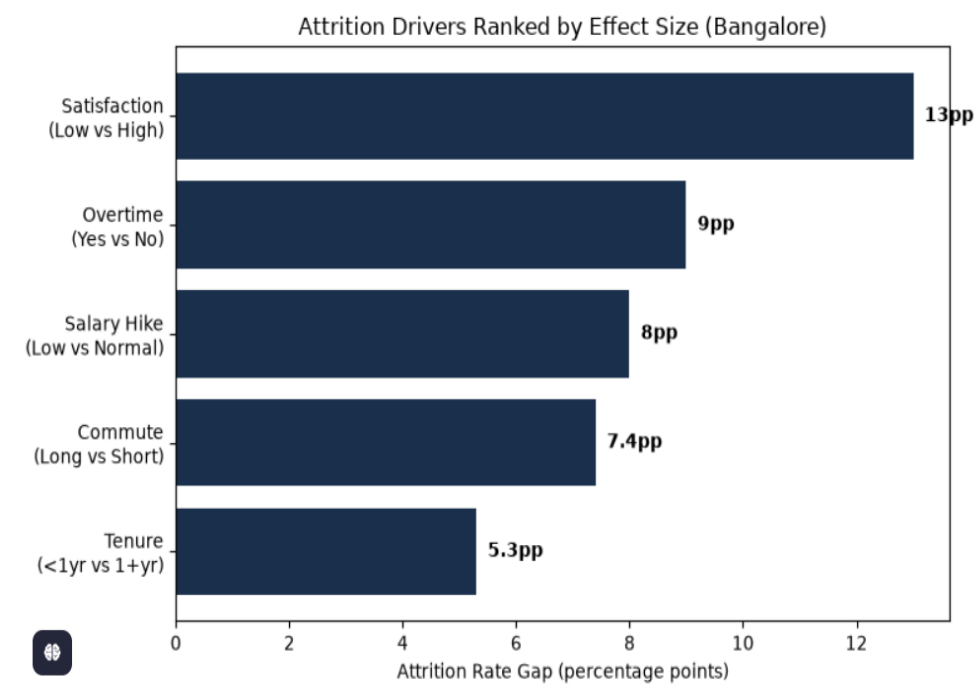
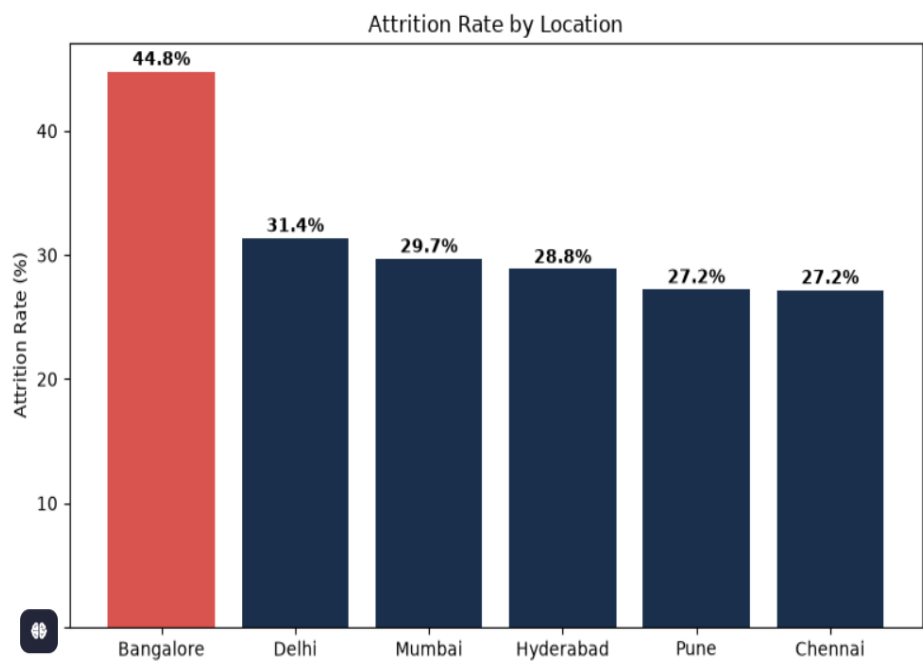
4. A residual, unexplained Bangalore-specific effect remains

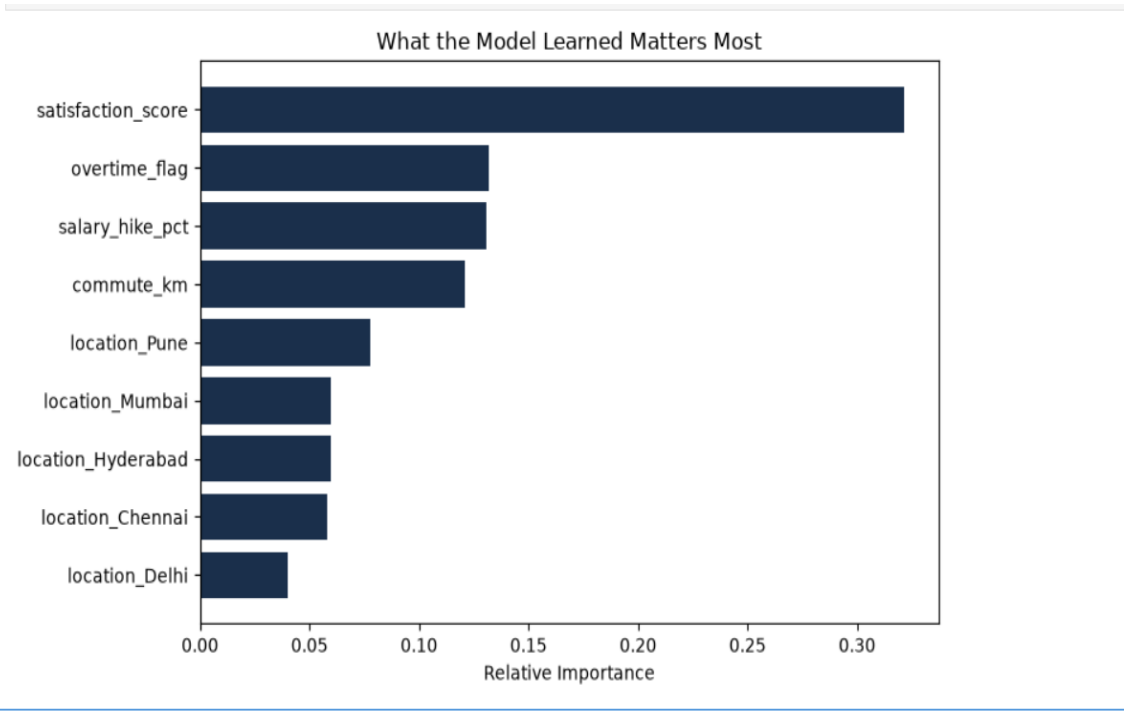
Even after accounting for satisfaction, overtime, salary hikes, and commute together, employees in Bangalore still showed elevated attrition risk relative to identical employees in other cities. This indicates a location-specific factor — such as management practices or local job market competition — not captured by the available data, and worth investigating directly through exit interviews or employee surveys.

5. Flight risk can be predicted, with a known and stated limitation

A classification model was built to flag employees likely to leave. After correcting for class imbalance, the model correctly identifies 58% of employees who go on to resign (recall), a major improvement over an initial uncorrected model that caught only 13%. This comes with a trade-off: roughly 57% of flagged employees will not actually leave. Given the low cost of a proactive manager check-in versus the high cost of an unanticipated resignation, this trade-off is considered acceptable for use as a screening tool.

Supporting Visuals





Recommendations

- Prioritize employee satisfaction initiatives in Bangalore first — the highest-leverage, most cost-effective intervention identified.
- Review and reduce mandatory overtime in Bangalore, which accounts for a disproportionate share of the company's total overtime hours.
- Reassess the salary hike floor for employees currently receiving below 6% — this group shows meaningfully elevated attrition risk.
- Use the predictive model as an early screening tool to prioritize proactive manager check-ins, not as a certainty about any individual employee.
- Investigate the residual Bangalore-specific effect through direct employee feedback, since it is not explained by the five factors measured.

Full methodology, data cleaning procedure, and model evaluation details are available in the project repository. Tools used: PostgreSQL for data cleaning and hypothesis testing, Python (pandas, scikit-learn) for predictive modeling.